



Global Earthquake Analytics Dashboard

Analysis of global earthquake data using USGS API, Python, Pandas, SQL and TiDB.

✓ TiDB Connected Successfully | 137,461 earthquake records loaded



Earthquake Overview

 Total Earthquakes

 Average Magnitude

 Maximum Magnit...

 Average Depth

 Tsunami Events

137,461

3.88

8.80

61.02 km

625



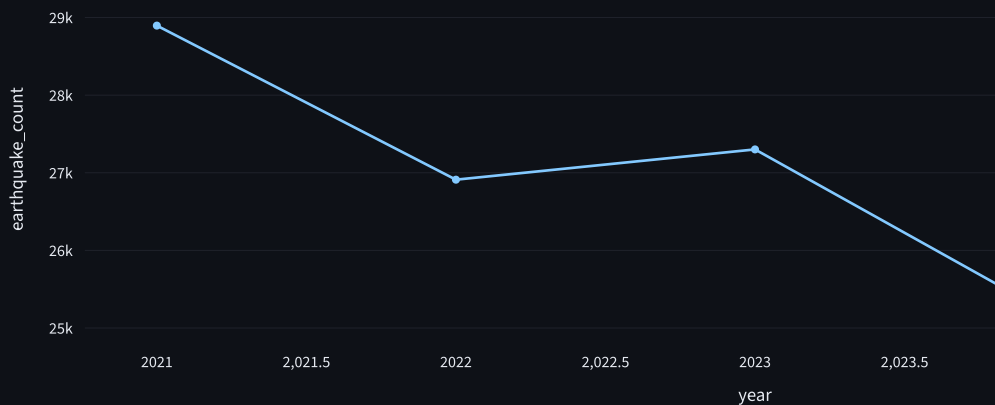
Global Earthquake Locations





Earthquake Trend by Year

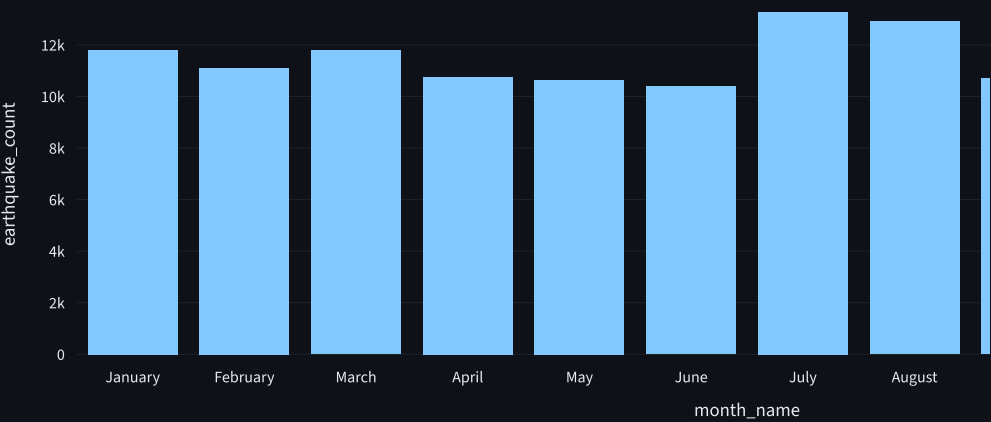
Earthquakes per Year





Earthquake Activity by Month

Earthquakes by Month

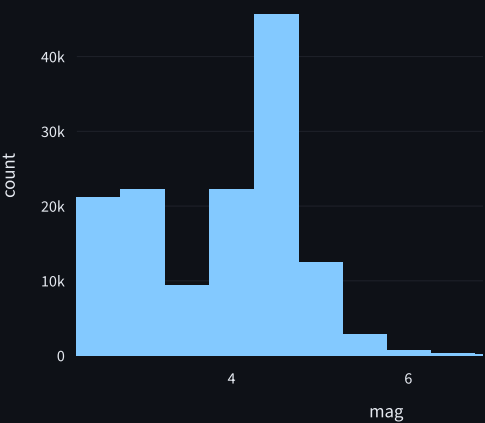


Magnitude Distribution

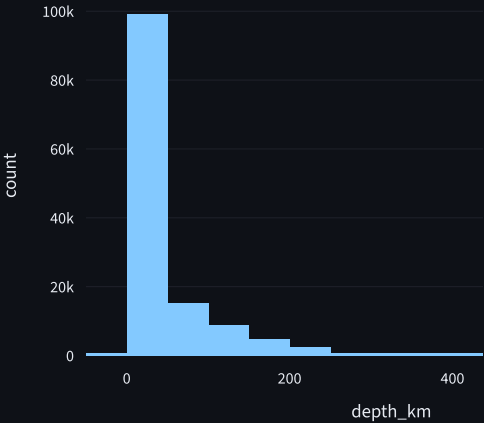


Depth Distribution

Magnitude Distribution



Depth Distribution



Average Magnitude by Magnitude Type

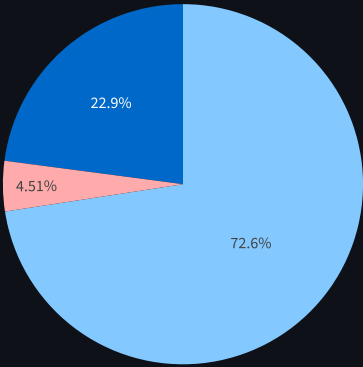
Average Magnitude by magType





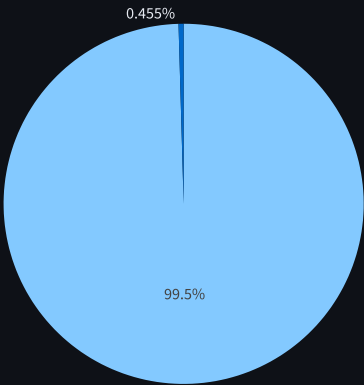
Earthquake Depth Categories

Earthquake Depth Categories



Tsunami Analysis

Tsunami vs Non-Tsunami

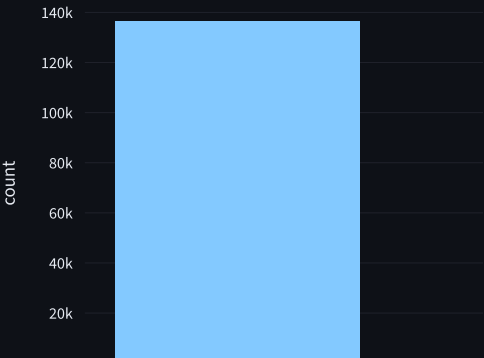


Reviewed vs Automatic

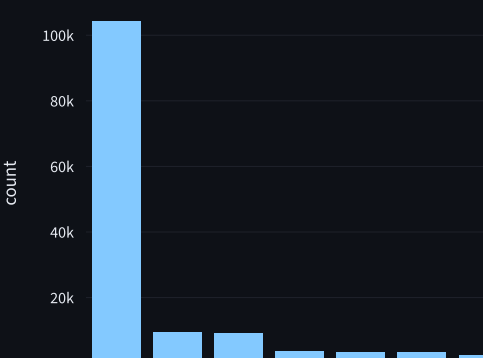


Top Reporting Networks

Earthquake Status



Top 10 Reporting Networks



reviewed

status

us

pr

ak

nc

hv

tx

c

network



Top 10 Strongest Earthquakes

time	place	mag	depth_km
2025-07-29 23:24:52	2025 Kamchatka Peninsula, Russia Earthquake	8.8	
2021-07-29 06:15:49	2021 Chignik, Alaska Earthquake	8.2	
2021-08-12 18:35:17	2021 South Sandwich Islands Earthquake	8.1	
2021-03-04 19:28:33	2021 Kermadec Islands, New Zealand Earthquake	8.1	
2025-09-18 18:58:15	140 km E of Petropavlovsk-Kamchatsky, Russia	7.8	
2023-02-06 01:17:34	Pazarcik earthquake, Kahramanmaras earthquake sequence	7.8	
2023-05-19 02:57:03	southeast of the Loyalty Islands	7.7	
2021-02-10 13:19:56	southeast of the Loyalty Islands	7.7	
2025-03-28 06:20:53	2025 Mandalay, Burma (Myanmar) Earthquake	7.7	
2023-12-02 14:37:04	19 km E of Gamut, Philippines	7.6	



Top 10 Deepest Earthquakes

time	place	mag	depth_km
2023-04-01 18:09:17	208 km ENE of Sola, Vanuatu	4	
2023-09-18 15:35:27	Vanuatu region	4.2	
2024-03-08 22:42:24	Fiji region	4.2	
2025-10-29 17:07:34	205 km ESE of Levuka, Fiji	4.8	
2021-08-01 22:12:46	283 km SE of Levuka, Fiji	4	
2025-05-10 02:03:17	299 km E of Levuka, Fiji	4.2	
2021-02-13 16:26:42	south of the Fiji Islands	4.5	
2022-06-01 18:41:01	279 km ESE of Labasa, Fiji	4.3	
2023-12-31 02:19:27	138 km NE of Sola, Vanuatu	4.1	
2022-11-09 09:51:04	south of the Fiji Islands	7	



Strong Shallow Earthquakes

Magnitude > 7.5 & Depth < 50 km

14

time	place	mag	depth_kr
2021-07-29 06:15:49	2021 Chignik, Alaska Earthquake	8.2	
2021-02-10 13:19:56	southeast of the Loyalty Islands	7.7	
2021-08-12 18:35:17	2021 South Sandwich Islands Earthquake	8.1	
2023-02-06 01:17:34	Pazarcik earthquake, Kahramanmaras earthquake sequence	7.8	
2023-05-19 02:57:03	southeast of the Loyalty Islands	7.7	
2025-07-29 23:24:52	2025 Kamchatka Peninsula, Russia Earthquake	8.8	
2025-10-10 20:29:20	Drake Passage	7.6	
2025-12-08 14:15:10	2025 Aomori Prefecture, Japan Earthquake	7.6	
2021-03-04 19:28:33	2021 Kermadec Islands, New Zealand Earthquake	8.1	
2022-09-19 18:05:08	35 km SSW of Aguililla, Mexico	7.6	

 Data Quality Metrics

High Station Coverage

105,635

Average Gap

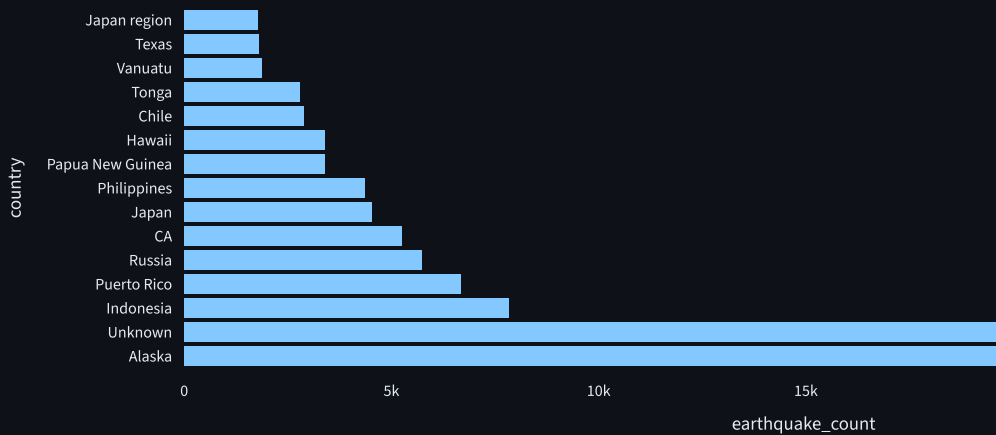
129.08

Average RMS

0.591

 Top Countries by Earthquake Frequency

Top 15 Countries



 Earthquake Dataset

Showing 137,461 filtered records

	id	time	updated	latitude	longitude	depth_km	mag	magType	place
0	ak02110e3pnu	2021-01-22 00:47:02	2021-04-13 17:05:49	55.0036	-157.6376	31.8	2.8	ml	7.0
1	ak02110ehjp6	2021-01-22 01:11:57	2021-04-13 17:07:01	60.1397	-153.4294	155.9	3.2	ml	5.0
2	ak02110hzoql	2021-01-22 07:00:08	2021-04-13 17:06:38	55.4983	-158.4934	2.3	2.5	ml	6.0
3	ak02110jg003	2021-01-22 09:44:15	2021-04-13 17:06:37	56.0673	-160.419	136.8	2.5	ml	4.0
4	ak02110oosqj	2021-01-22 18:25:01	2021-04-13 17:04:18	63.0042	-150.5144	93.3	2.6	ml	5.0
5	ak02110qyith	2021-01-22 22:06:29	2021-04-13 17:05:48	66.2993	-157.2101	5	2.8	ml	6.0
6	ak021129yxh5	2021-01-23 14:46:10	2021-04-13 17:07:14	62.1493	-151.165	74.5	2.5	ml	2.0
7	ak02112d03wz	2021-01-23 19:55:49	2021-04-13 17:07:15	61.1555	-150.5539	49.4	2.6	ml	2.0
8	ak02112erqia	2021-01-23 22:52:35	2022-08-09 02:36:37	60.1077	-152.9254	109	2.7	ml	6.0
9	ak02113w3sj0	2021-01-24 12:10:56	2021-04-13 17:06:23	58.6798	-155.0667	137.3	2.7	ml	8.0
10	ak021142pa6g	2021-01-24 23:18:58	2021-04-13 17:05:48	59.1296	-137.966	3.3	3.2	ml	1.0
11	ak02115hueu8	2021-01-25 09:08:56	2021-04-13 17:07:32	65.6469	-145.1672	15.6	3	ml	1.0
12	ak02115injdf	2021-01-25 10:44:45	2021-04-13 17:05:51	62.5125	-149.898	13.9	2.9	ml	1.0

 Download Filtered Dataset